Artificial Intelligence: From Biology to Industry



Copyright:Prof. Dr. Marzuki Khalid (Center for AI and Robotic, UTM). Keynote: International Conference on AI in Engineering and Technology (2002)

Scope of Presentation

- What is Artificial Intelligence?
- Models of Biological Intelligence

  (The Human Brain)
- Examples of AI Applications in Industry

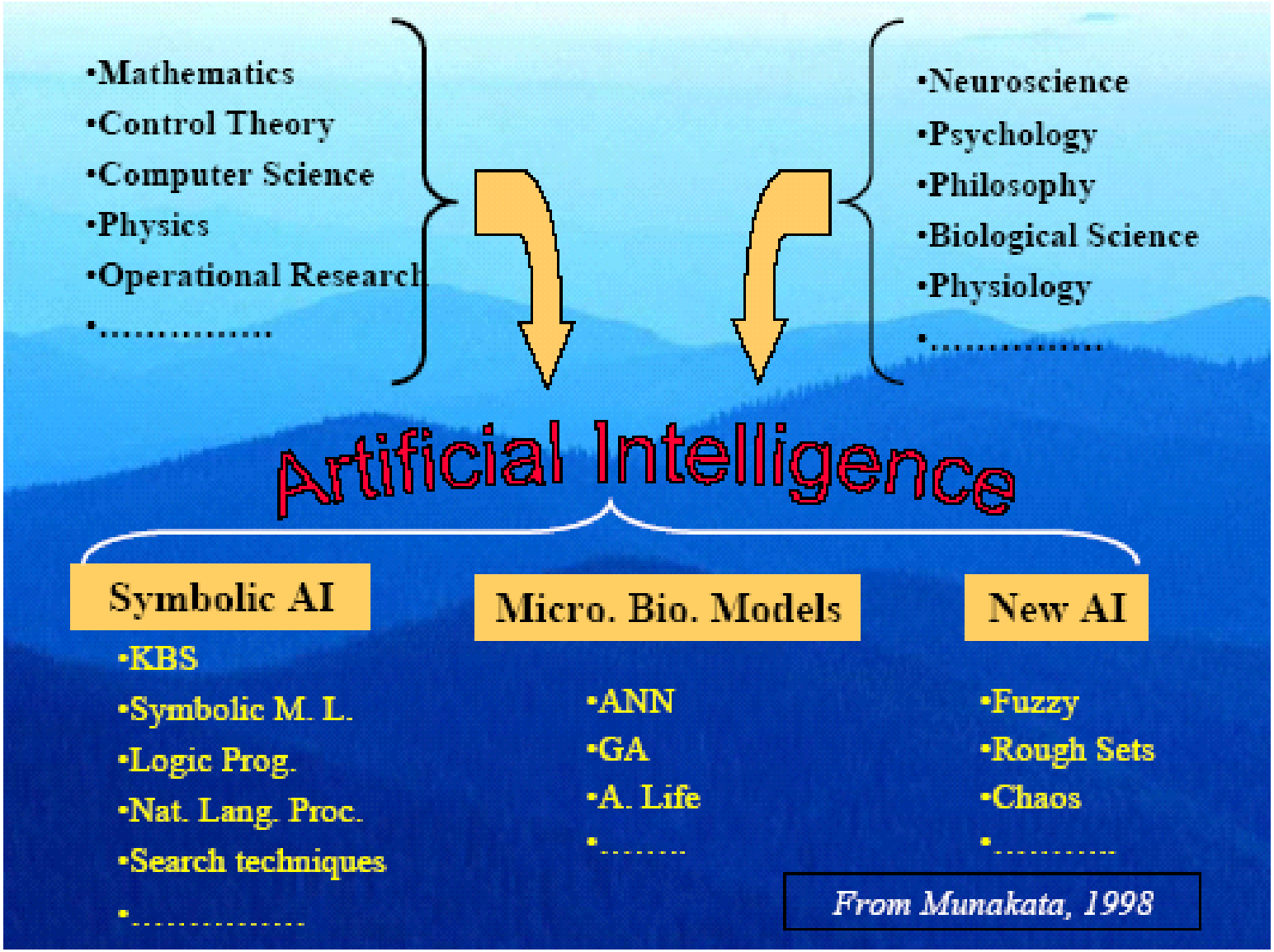


What is Artificial Intelligence?

Intelligence is defined as clever, knowledge, quickness in learning, information, etc.



AI is a "tool" that has been developed to imitate human intelligence and decision-making functions, providing basic reasoning and other human characteristics.


•Mathematics
•Control Theory
•Computer Science
•Physics
•Operational Research
•.....
•Neuroscience
•Psychology
•Philosophy
•Biological Science
•Physiology
•.....
Artificial Intelligence
Symbolic AI
•KBS
•Symbolic M. L.
•Logic Prog.
•Nat. Lang. Proc.
•Search techniques
•.....
Micro. Bio. Models
•ANN
•GA
•A. Life
•.....
New AI
•Fuzzy
•Rough Sets
•Chaos
•.....
From Munakata, 1998

Terminologies associated with

Artificial Intelligence:

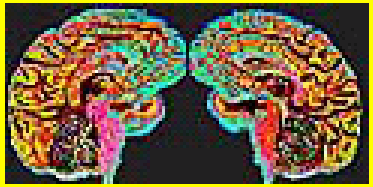


- **Cybernetics**
- **Systems neurobiology**
- **Computational neuroscience**
  (Sejnowski and Churchland)
- **Connectionism**
  (Barto et.al.)
- **Parallel distributed processing or PDP**
  (Rumelhart and McClelland)
- **Soft Computing**
  (Zadeh)



Scope of Presentation

- What is Artificial Intelligence?
- Models of Biological Intelligence (The Human Brain)



- Examples of AI Applications in Industry

SUMMARY OF FACTS ABOUT THE HUMAN BRAIN

- **Has self-organizing or self-learning capability. It can easily adjust to new environment by "learning" usually from experience or from a knowledge of its surrounding.**
- **Robust and fault tolerant.**
- **Parallel distributed architecture.**
- **Can deal with information that is fuzzy, probabilistic, noisy, or even inconsistent (nonlinear).**
- **Small, compact and dissipates very little power.**

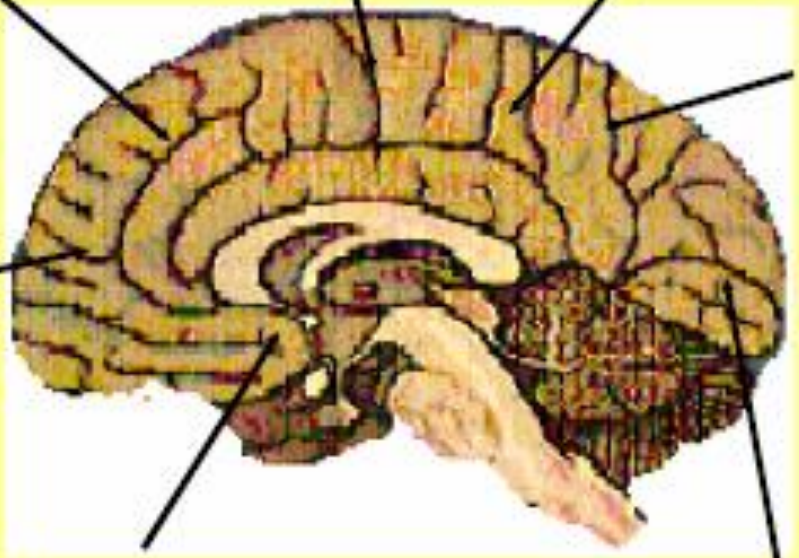
- A typical adult human brain weighs about 1.3kg (3lb).
- About 0.85 liters (1.8 pints) of blood travel through the brain every minute.
- A human brain contains about fifteen billion cells.



Motor Cortex
- Fine Motor
- Fine Tuning

Sensory Cortex
- Skill Sensation (Pressure)
- Taste

Frontal Lobe
- Critical Thinking
- Problem Solving



Parietal Lobe
- Touch
- Proprioception

Prefrontal Cortex
- Planning
- Rehearsal

Temporal Lobe
- Hearing
- Speech

Occipital Lobe
- Vision

LEVELS IN THE NERVOUS SYSTEMS

The *nervous systems* can be studied by breaking them up into a number of levels *(Churchland and Sejnowski).*



Thus, 3 approaches:

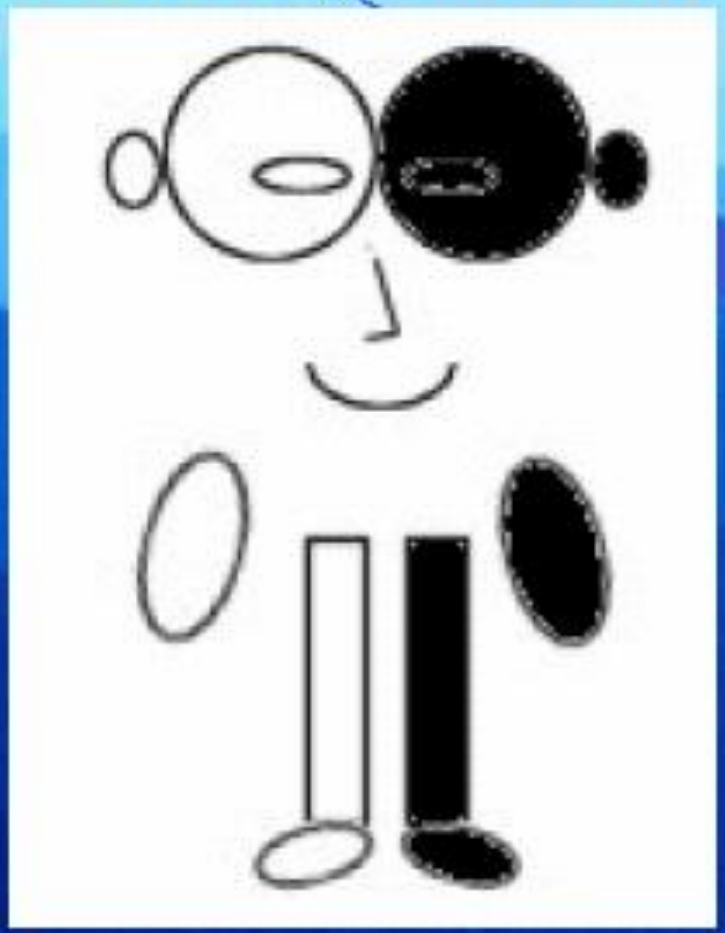
- Levels of Organizations
- Levels of Analysis
- Levels of Processing

Let's see how your brain works

Right

Left






Balancing the use of your Left and Right Brain can make you perform better

LEARNING SUCCESS INDICATOR

- Unconscious Incompetence
- Conscious Incompetence
- Conscious Competence
- Unconscious Competence

NOVICE



NATURAL

Greg Barcelon

Creating a World Class Team

NOVICE

- Novice – Rules
- Advanced Beginner – Rules + Context
- Competent – Context + Rules
- Proficient – Patterns
- Expert – Part of Life (Natural)



NATURAL



Dreyfuss

PSYCHOLOGY



Neuro Linguistic Programming

John Grinder, Richard Bandler

Experts Creating New Experts

Learning Model

(1) Belief System of the person

(2) Mental syntax: sequencing of their thought

(3) Physiology (how do we dress?, how do we talk?, etc.

Psychology

Neuro-Linguistic Programming



SCOTTISH SHEEP SHOCKER!

Look mom -- no mom!



Scientists suddenly have the capability to create countless

Medicine

Cloning of Humans/Animals from Cells

Artificial Intelligence

Machine Intelligence





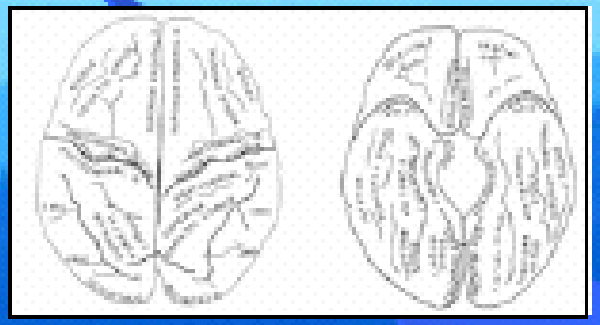
Arc-Welding Robots

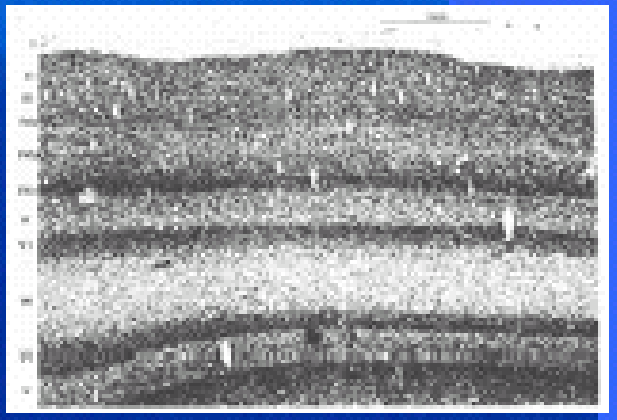
LEVELS OF ORGANIZATION



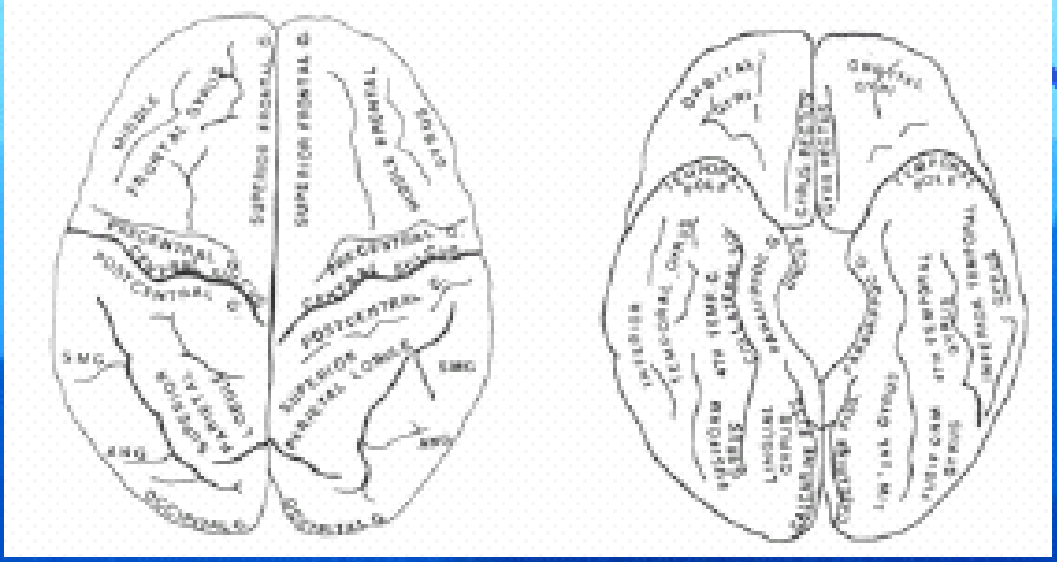
The Brain:

- Systems
- Maps
- Layers
- Networks
- Neurons
- Synapses
- Molecules





MAPS IN THE BRAIN

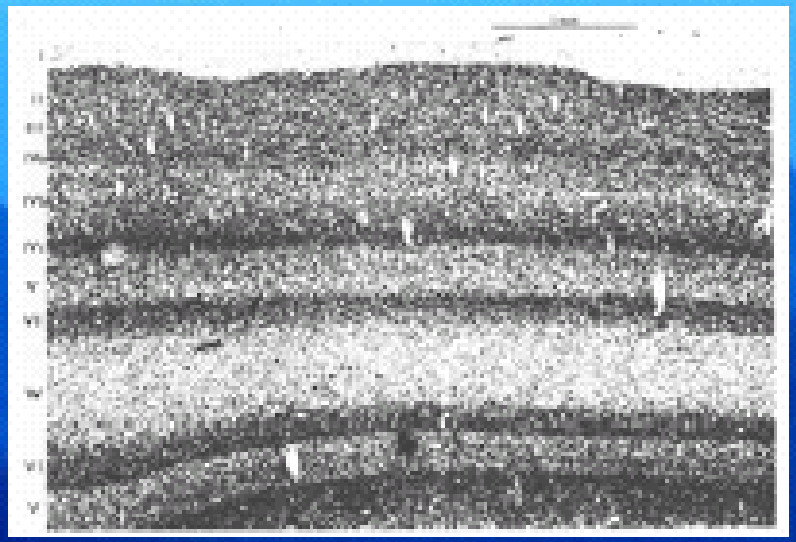




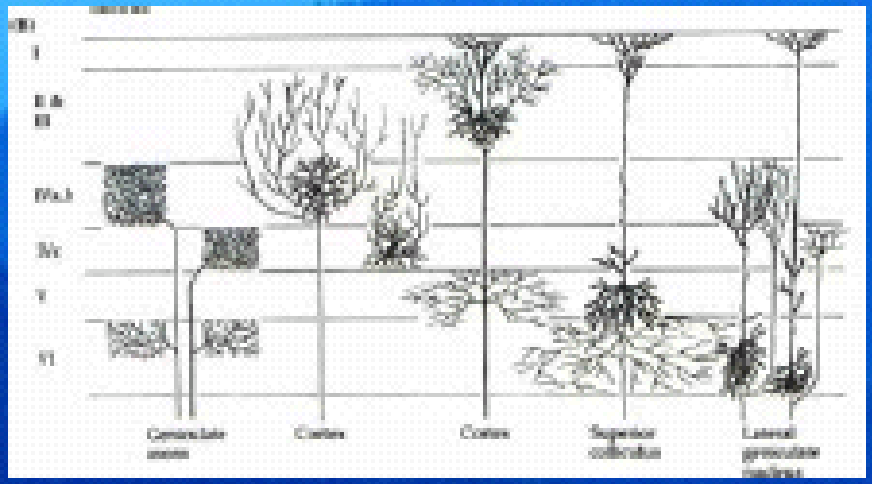
- **For example, neurons in visual areas of cortex are arranged topographically, in the sense that adjacent neurons have adjacent visual receptive fields and collectively they constitute a map of the retina.**

LAYERS IN THE BRAIN

- THE BRAIN IS ALSO DIVIDED INTO MANY LAYERS, EACH LAYER SERVES DIFFERENT PURPOSES.



Cross-section through a monkey striate cortex

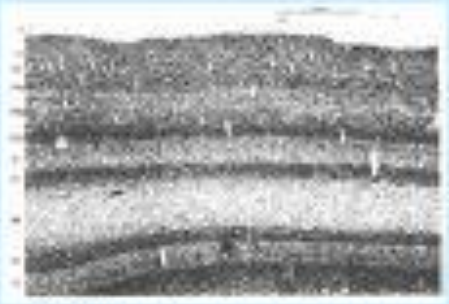


Schematic diagram of cortical connections in the cat.

Copying from Biological Systems

COMPUTER ARCHITECTURES

- The Motherboard: from single layer to multilayer

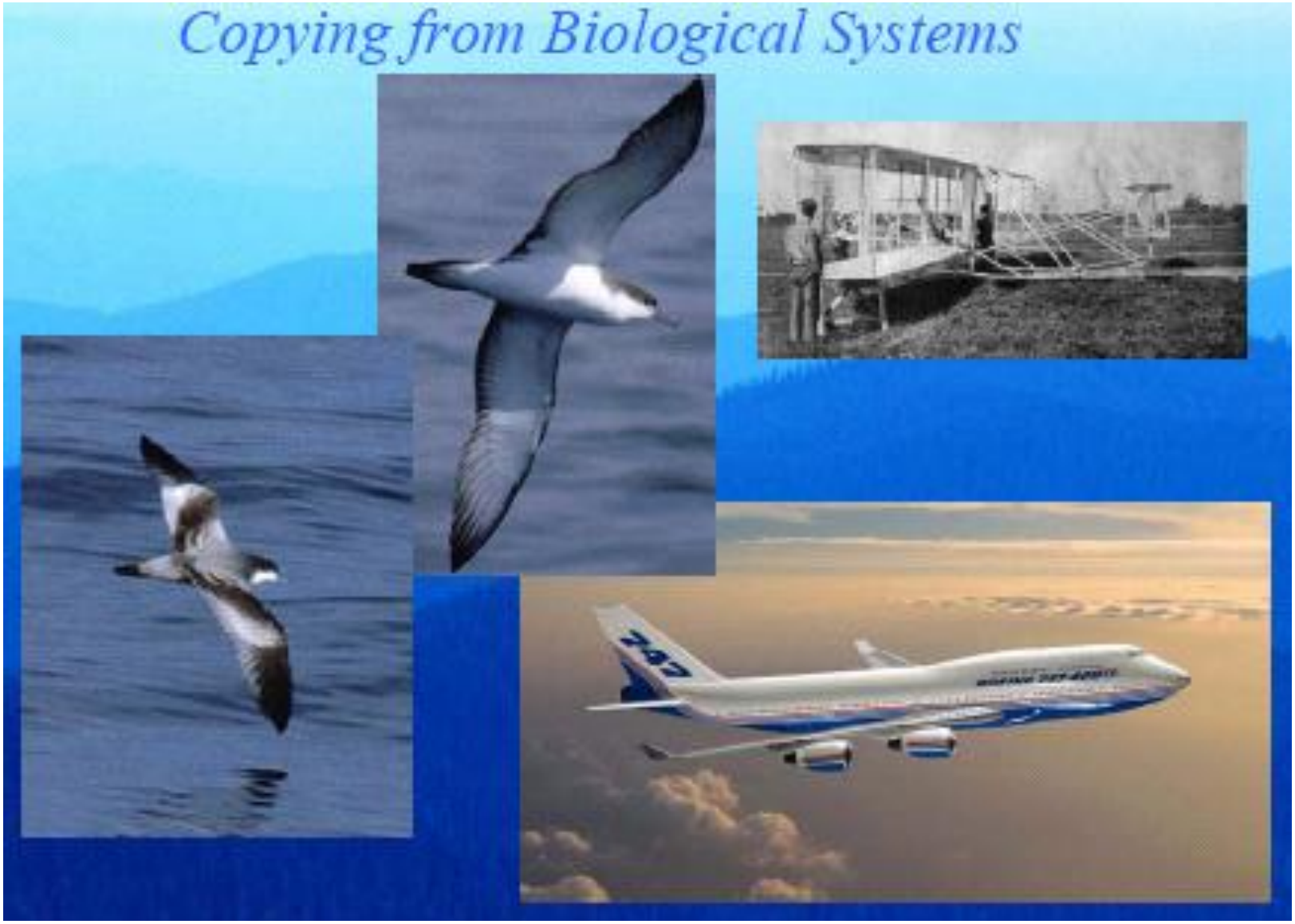








Copying from Biological Systems



Copying from Biological Systems



In a study by T. Yoshikawa (Kyoto University), the design of the optimal length of robotic manipulators should follow that of the human arm





Automated Disassembly Equipment for Mobile Platforms

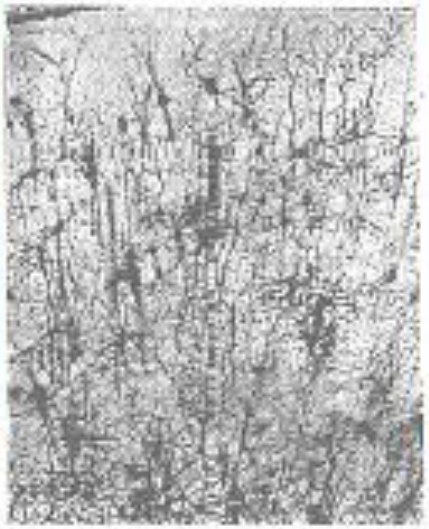
Copying from Biological Systems

From Thumb Print → Bar Code

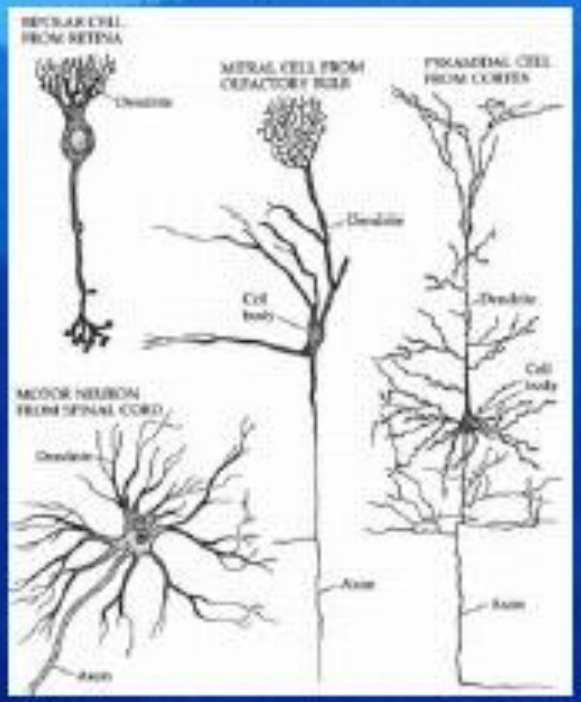


Everyone, including identical twins, has a unique fingerprint. In other words, people's identities are coded at their fingertips. This coding system may also be compared to the barcode system that is used today.

THE NEURON




Neuron


BIPOLAR CELL FROM RETINA
MITRAL CELL FROM OLFACTORY BULB
PYRAMIDAL CELL FROM CORTEX
Dendrite
Cell body
Dendrite
Dendrite
Cell body
MOTOR NEURON FROM SPINAL CORD
Dendrite
Axon
Axon
Axon

From Biological Neuron to Artificial Neuron



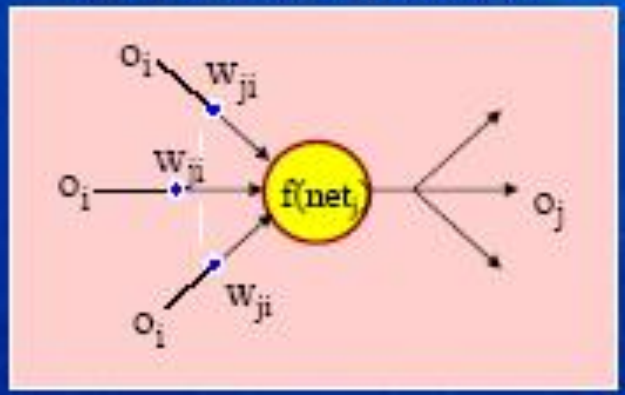




The first artificial neuron was developed by McCulloch and Pitts (1943)

"A logical calculus of the ideas immanent in nervous activity", Bulletin of Math. Biophysics, Vol.5, pp115-133, 1943

AN ARTIFICIAL NEURON



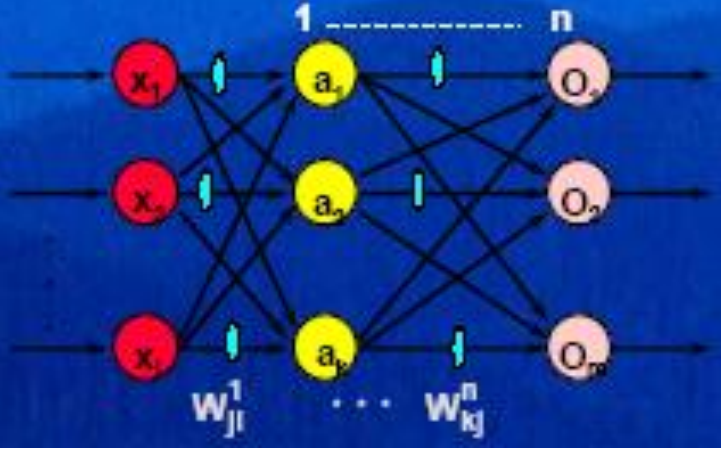
Biological Network





Artificial Neural Network

An artificial neural network is a universal function approximator. Weights W_{ij} are "learned" to fit any function.





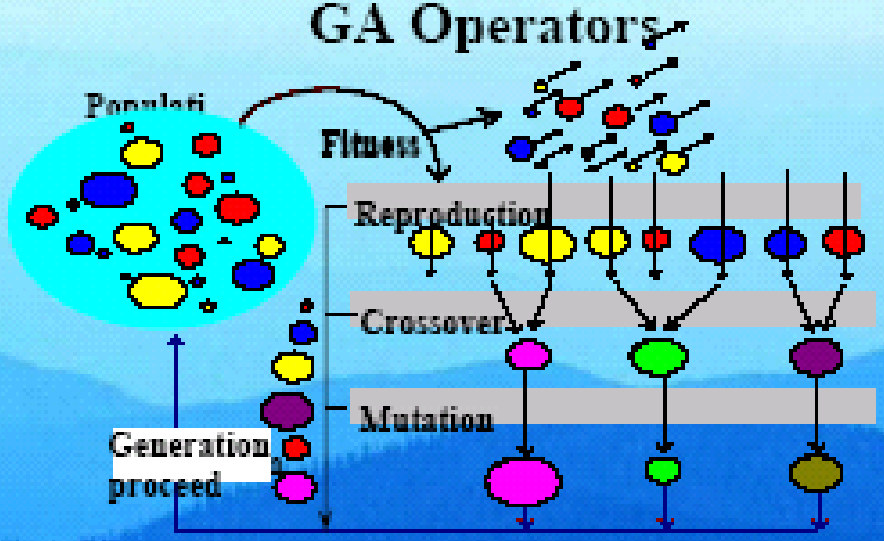
A Summary of Artificial Neural Networks

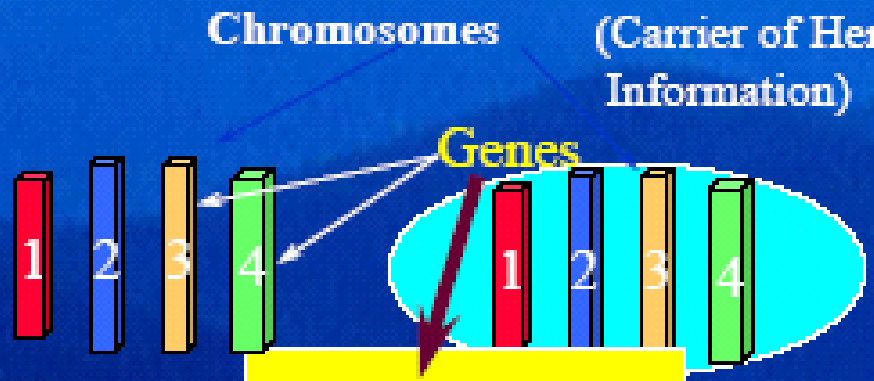
- Based on the biological system
- An information processing system
- Neurons
- Interconnections
- Activations function.
- Associated weights
- A learning rule.
- Training patterns

Some ANN Paradigms:

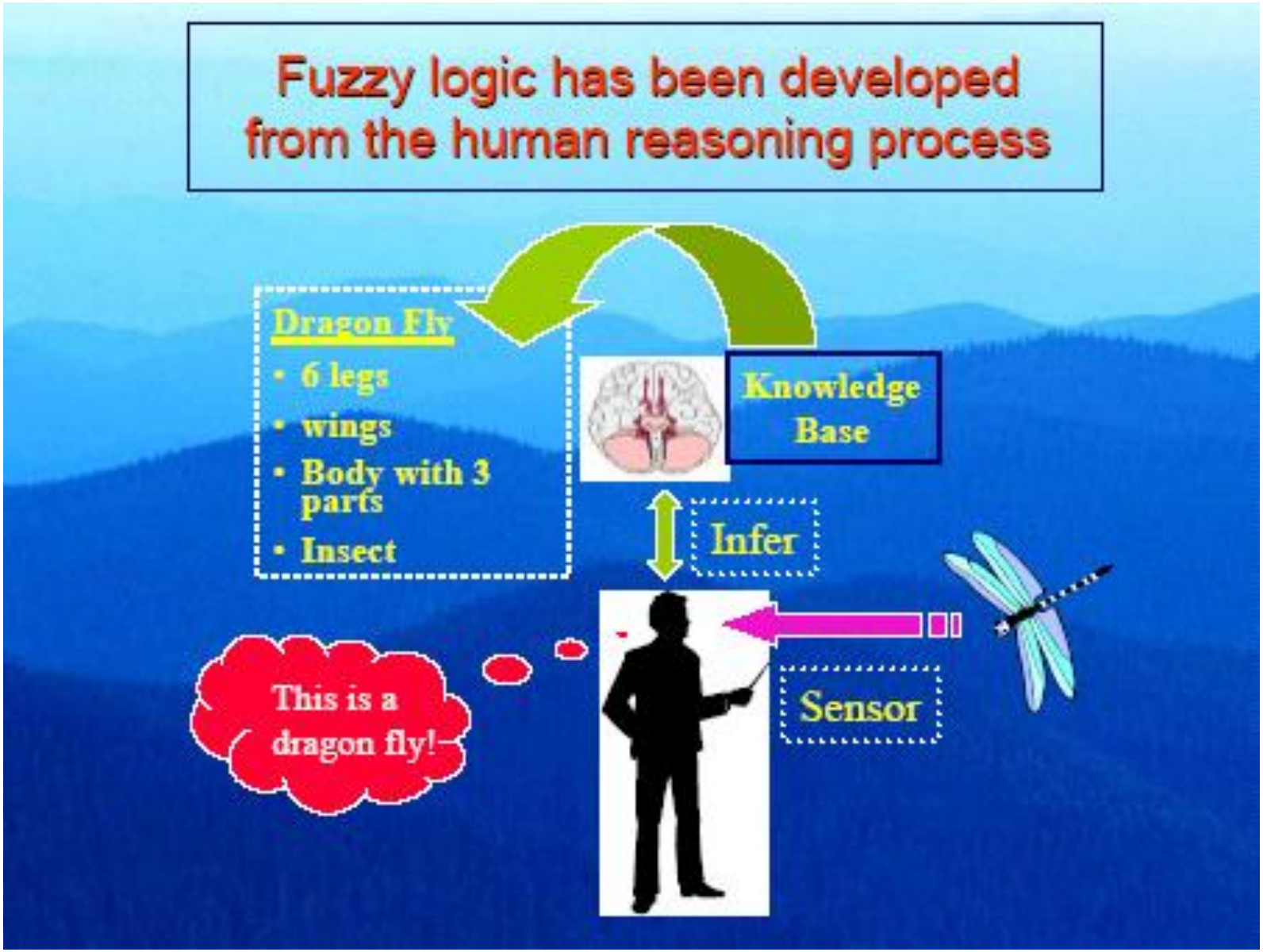
- Kohonen SOM
- BP-based Multilayer NN
- ART NN
- GRNN
- BAM
- Hopfield NN
- Competitive Learning
- PNN

Genetic Algorithms have been derived from the human reproduction process





Features, Characters:
Eg. Dark Hair,
Brown Eyes, Tall

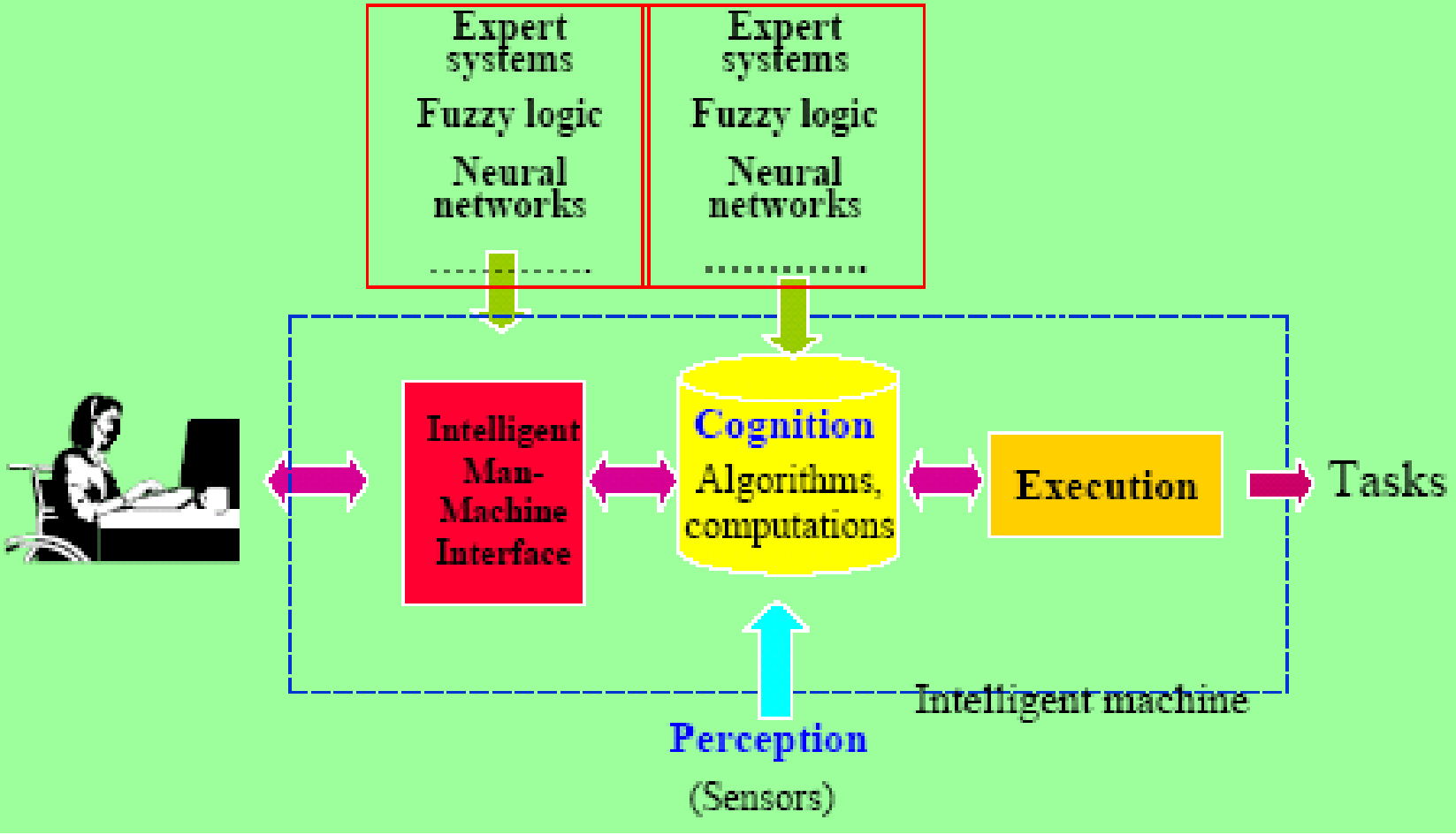

Fuzzy logic has been developed from the human reasoning process
Dragon Fly
• 6 legs
• wings
• Body with 3 parts
• Insect
Knowledge Base
Infer
Sensor
This is a dragon fly!

- TABLE OF LEFT vs. RIGHT BRAIN FUNCTIONS
- http://acceleratedlearningmethods.com/brain-functions.html

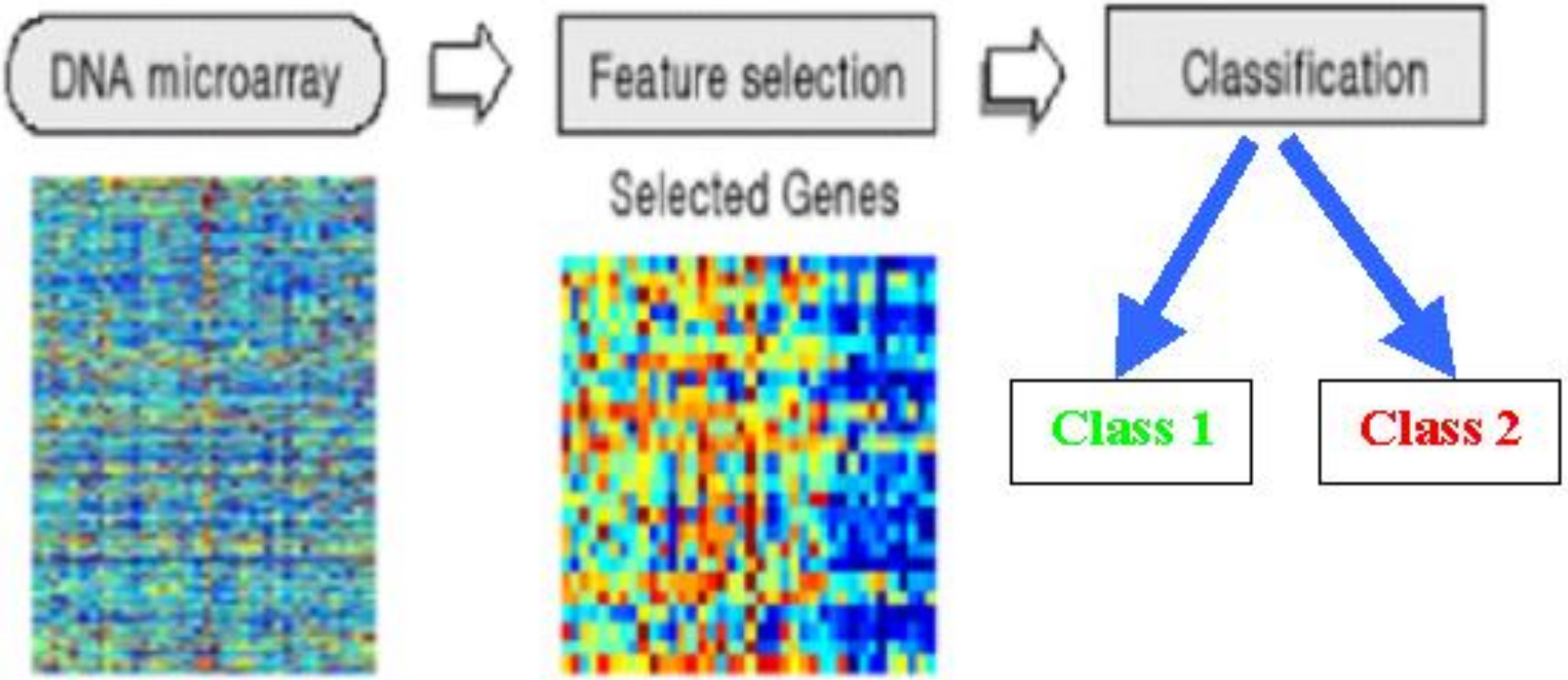
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- Examples of AI Applications in Industry

AI Applications in Industry

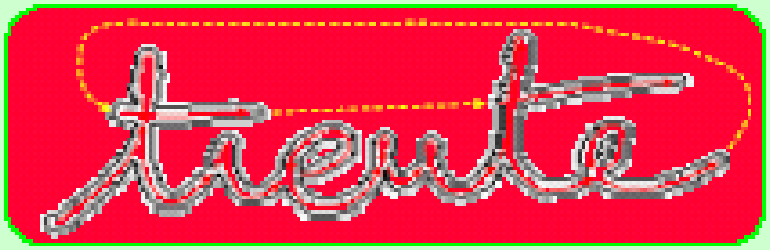


Model of Cancer Disease Classification from Gene Expression Data



Research in Handwriting Recognition

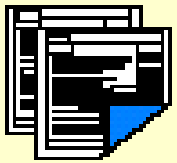
(with Ecole Polytechnique Univ. de Nantes, France)



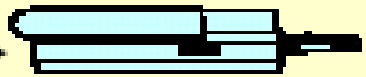


- Online Recognition
  - Cursive
  - PDAs
  - Signature Verification
- Offline Recognition
  - Bank Cheque Reading
  - Mail Reading
  - Form Recognition

Automated Data Entry System (AUDESY)



Hand-filled forms



Forms are scanned

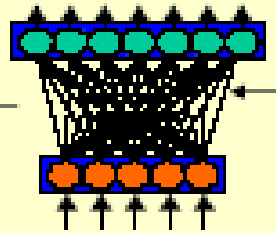


Images of forms are collected

Skew detection/ Alignment

Image processing

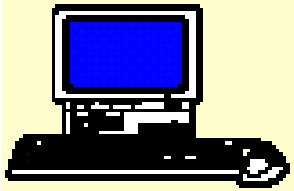
Feature extraction (GF)



Character recognition by Neural network



Verification



Storage into database

Target Applications

- National Registration Dept.
- Income Tax
- Immigration
- Custom and Excise
- Hospitals/Banks
- Statistics Dept

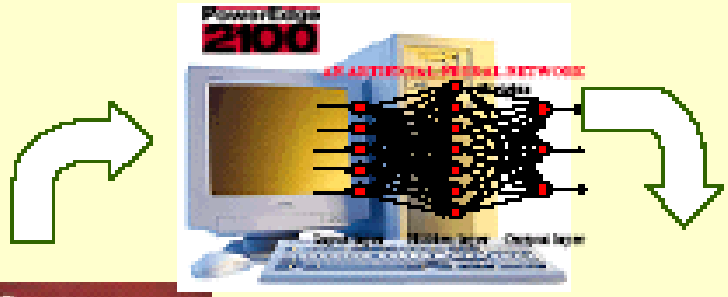


VISIONPLATE - Vehicle License Plate Recognition

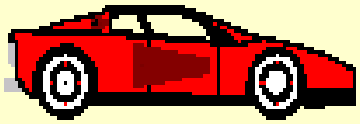
Image Processing

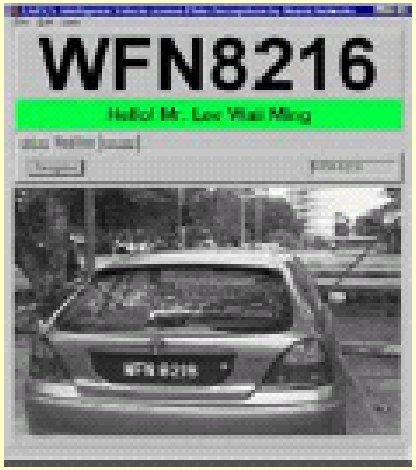
Feature Extraction

Neural Networks

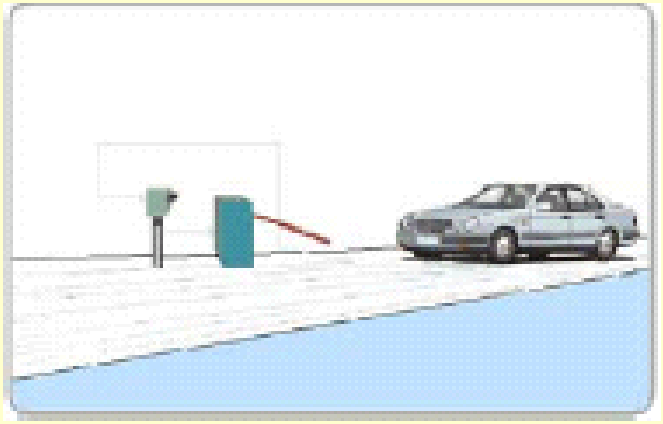




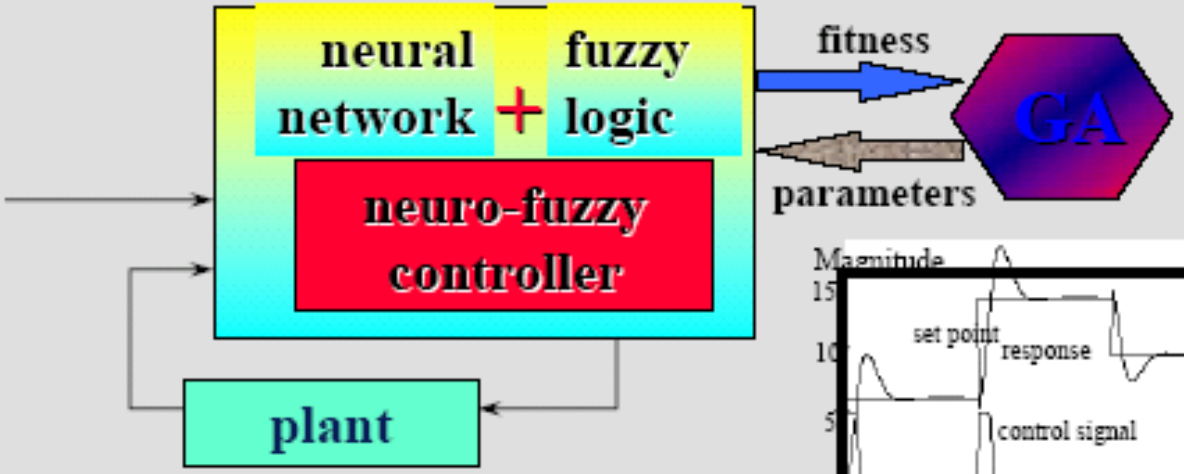


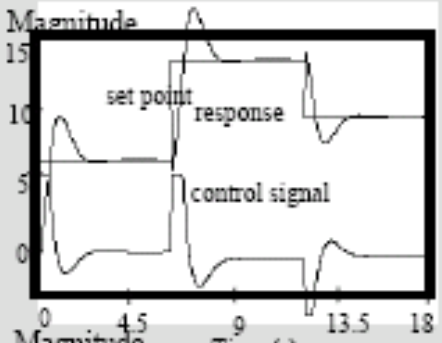




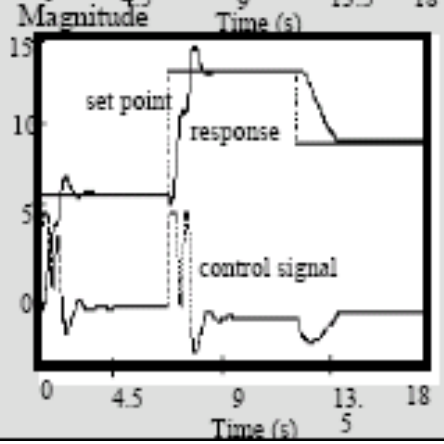


Neuro-Fuzzy Control Systems





GA-Tuned PID

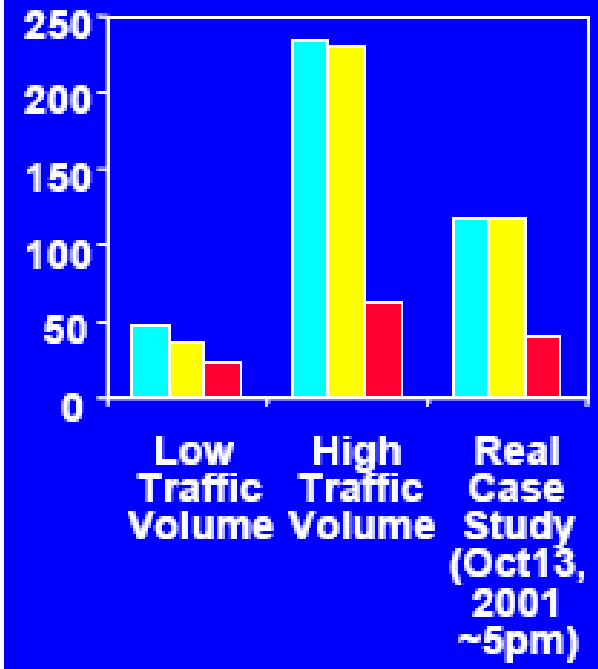


Neuro-Fuzzy Controller

Tuning by the GA:

- Centers & widths of membership
- Interconnected weights
- I/O Scaling factors

Adaptive Control of Traffic Lights using Fuzzy Inference



- Preset Cycle Time
- Vehicle Actuated
- Proposed Fuzzy Controller





Internet based Control and Virtual Laboratory



Client Computer

Camera



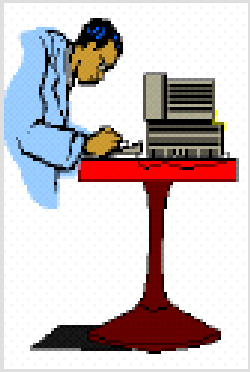


Coupled
-Tank

TCP/IP
communication

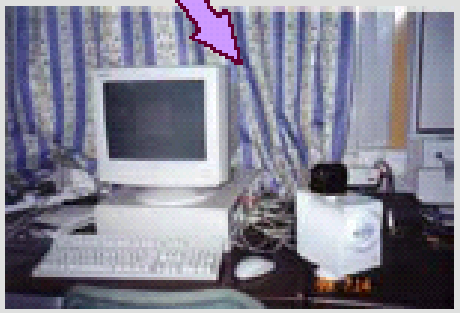
Internet

Server Computers

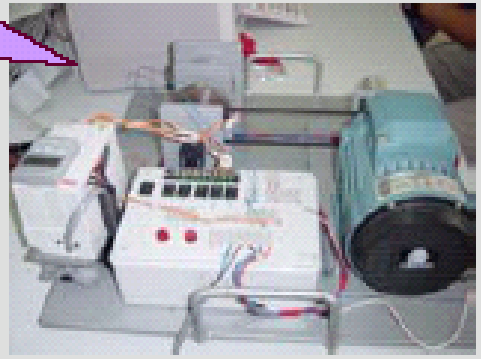




Client Computer



Client Computer



AC
Motor

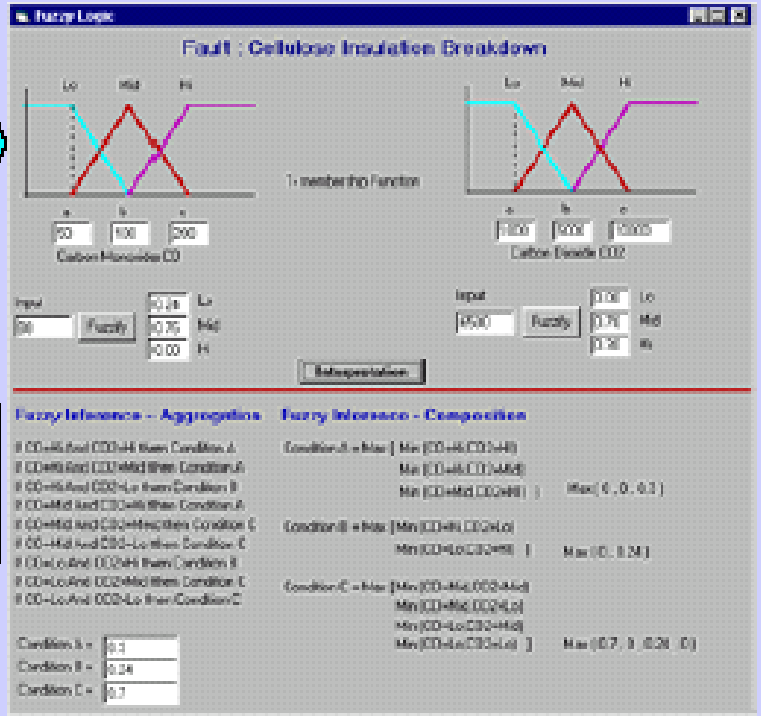
- access through web browser
- operator control
- online monitoring

Intelligent Transformer Diagnosis using Fuzzy Inference





Automatic Interpretation using Fuzzy Logic



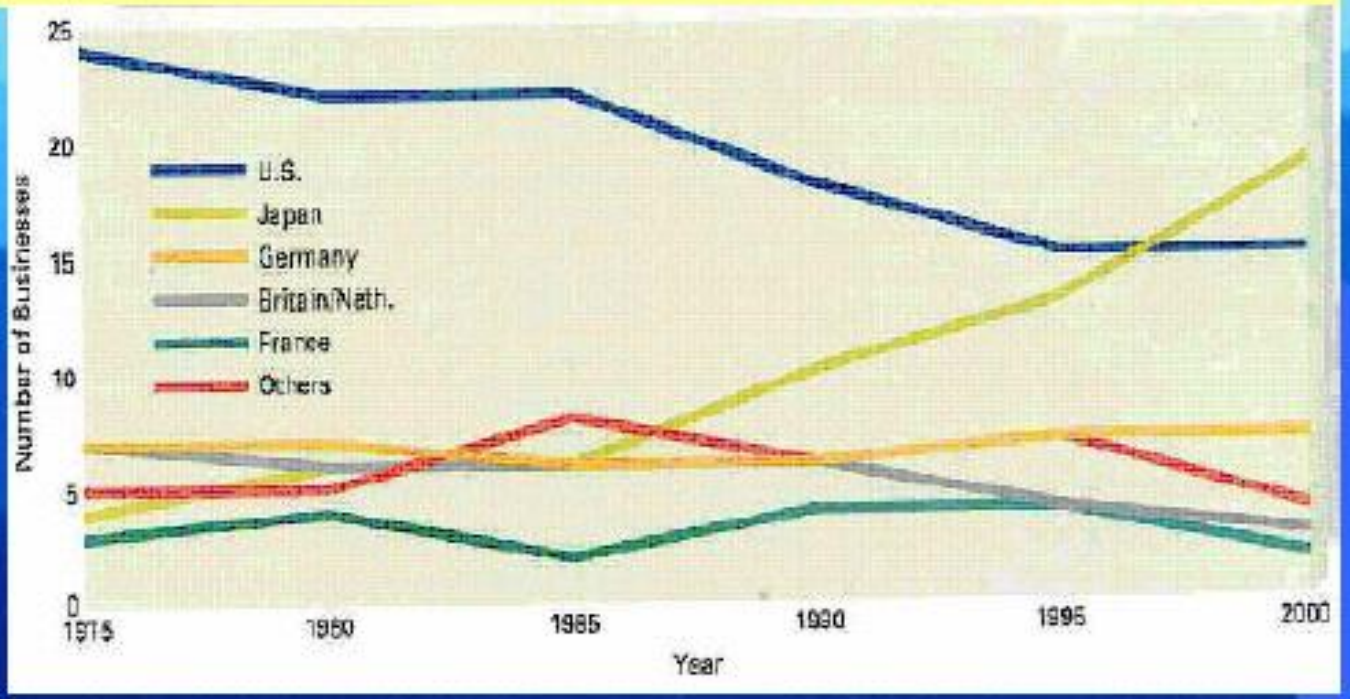
FUZZY INTERPRETATION

Transformer Condition

- ☺ Good
- 😐 Normal
- ☹ Bad

World Industrial Leaders by Country
(1975-2000)

(MIS for the Info Age)



In 1994 alone Japan sold US$34 billion worth of consumer products using fuzzy logic technology

Successful Applications of AI Technology

Vacuum Cleaner

Camera

Rice Cooker

Television



Air Conditioner

Washing Machine

Refrigerator

Stock Trading

Camcorder Stabilizer

Sendai Subway Control

ABS Braking System

Elevator Group Control

Cement Kiln Control



A PRIME ACHIEVEMENT

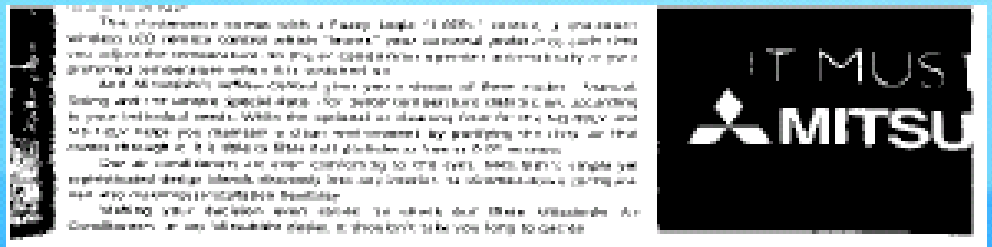
WAY ABOVE YOUR EXPECTATIONS

Intelligent Perceptions

A **Fuzzy Logic System** in the SEi model assists the driver's perception and judgement by automatically selecting the shift pattern best suited to road conditions based on several sets of built-in data like road winding conditions and gradients, and frequency of brake applications.

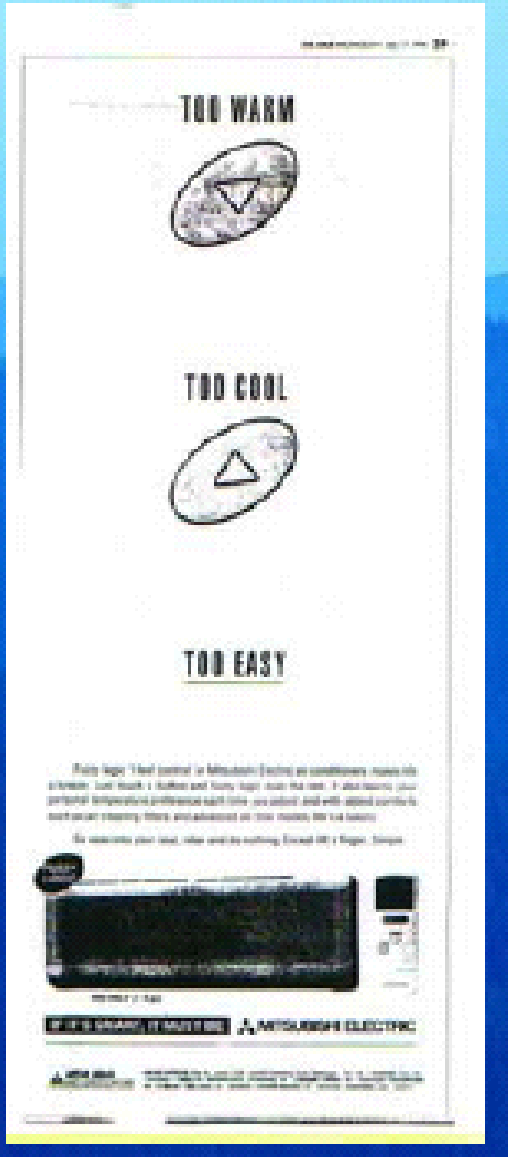
Fuzzy Logic Technology is becoming more affordable

Advertisement in Malaysian Newspapers

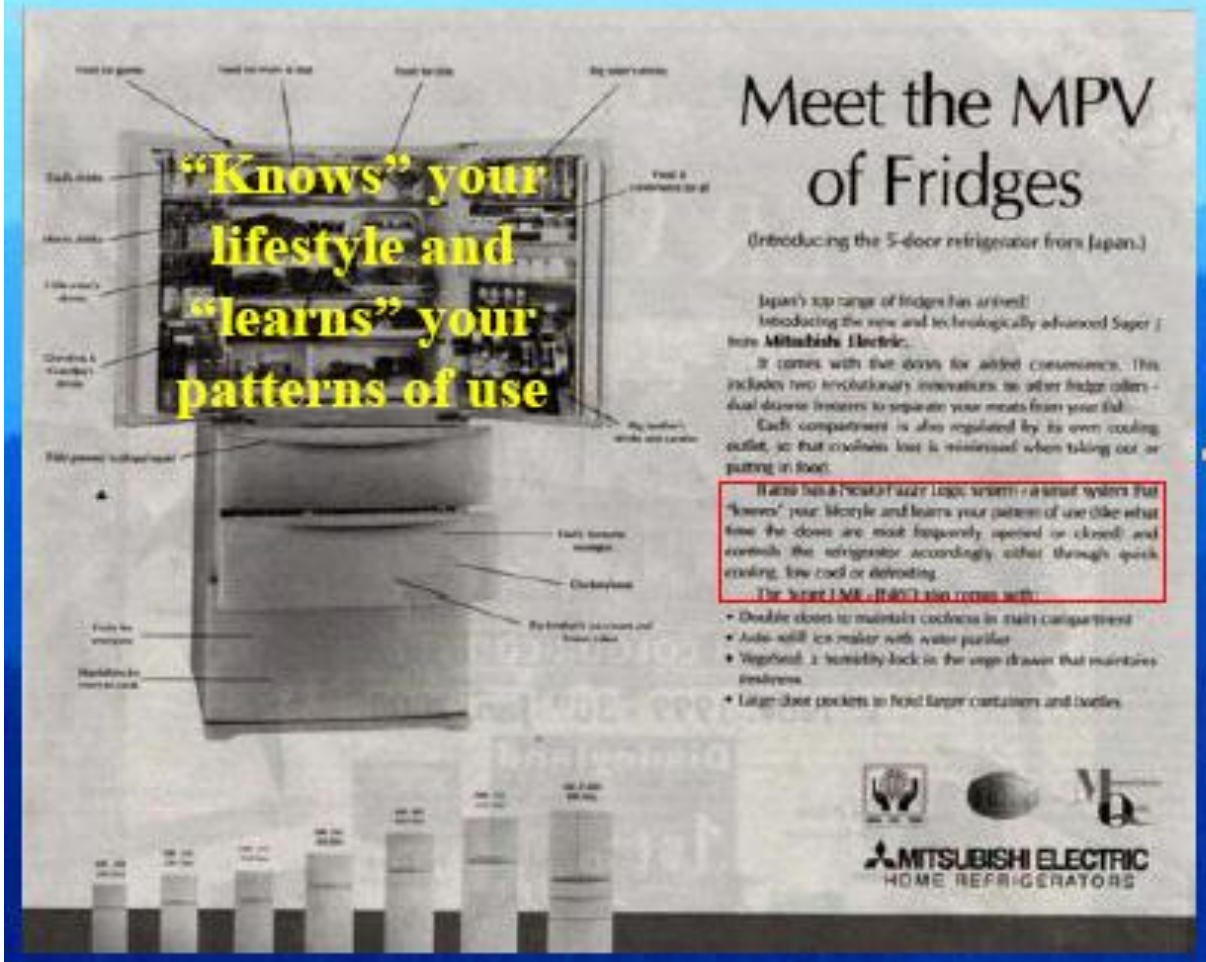




With *affordable*
Fuzzy Logic technology,
your choice of air conditioners
should be *simple*.



Advertisement in Malaysian Newspapers



It also has a Neuro-Fuzzy Logic System- a smart system that "knows"your lifestyle and learns your pattern of use(like what time the doors are most frequently opened or closed) and controls the refrigerator accordingly either through quick cooling, low cool or defrosting.

Advertisement in Malaysian Newspapers



This Refrigerator has a Neuro-Fuzzy Control System (For Door Cooling and Super-Cooling and Freezing)


Made in Japan
(based on a video on innovations and the need for change)
1960s
Junk
Cheap
Poor Quality
Copies
Low Technology
Imitation
1980s
Excellent Quality
Expensive
Leadership
Balance of Payment
High Technology
Innovations

CONCLUSION

- Many AI techniques are derived from Biological phenomena
- New techniques are required for improvement, thus, the need to further study other aspects of the biological models
- Application of AI in industry does improve model performance which can provide competitive advantage